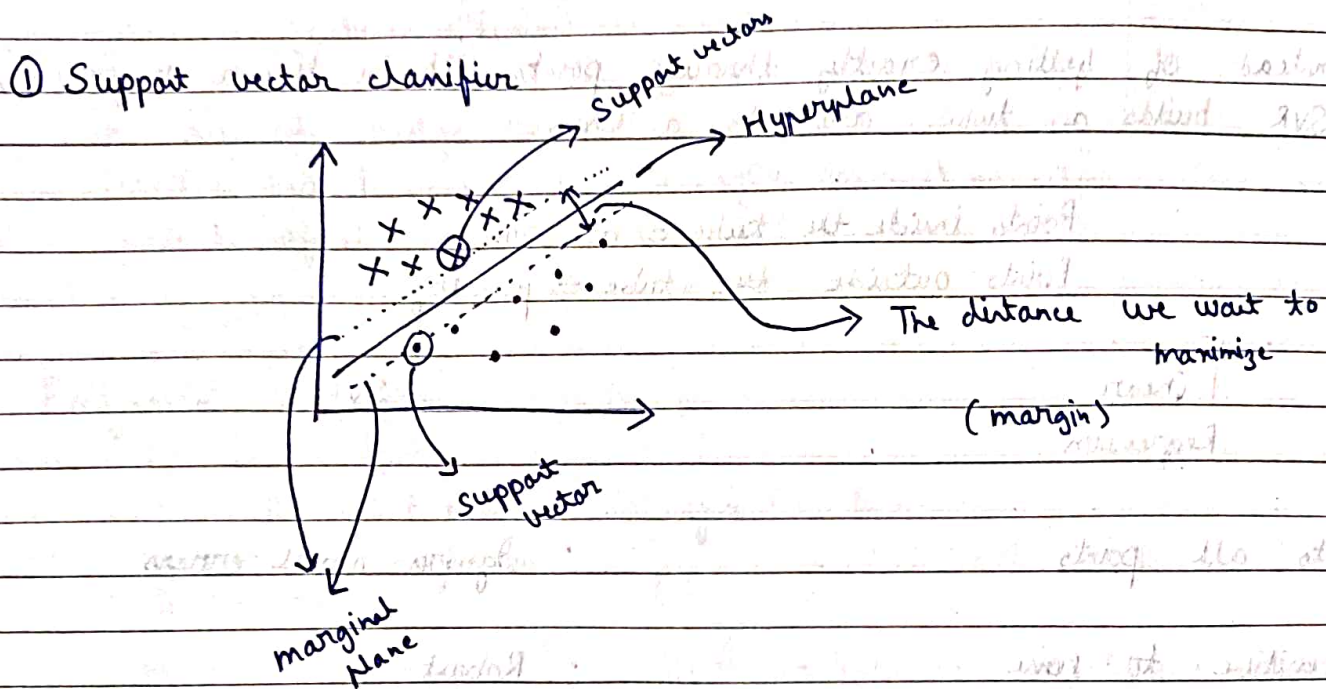


⑦ Support Vector Classification Machine

Tasks which can be solved using SVM (Support vector machines):

- ① Classification
- ② Regression



SVM is a classifier that separates data into classes by drawing the best possible boundary b/w them.

What makes a line 'best' in SVM?

SVM chooses the line that keeps maximum distance from the nearest data points of both classes.

This distance is called margin.

The closest points to the boundary are called support vectors.

SVM maximizes distance so small changes in data do not change the classification. So that a small push from one side doesn't make the point cross over to the different side of the boundary.

* In SVC the closest points to the wall from both sides are called support vectors.